What is in a Genre?

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Project Type: analysis of a dataset
Dataset: Spotify Music Dataset (Kaggle)

We propose using the **Spotify Music Dataset** from Kaggle to examine what defines musical genres based on measurable audio features such as danceability, energy, tempo, and valence. The project will aim to **predict a track's genre** using these features and compare the predicted genre to the labeled genre in the dataset.

We will compare multiple modeling approaches, including **supervised classification** (e.g., K-Nearest Neighbors) and **unsupervised clustering** (e.g., K-Means), to assess how well different techniques capture genre structure. Additionally, we plan to explore a third approach by passing the same feature data to a **large language model (LLM)** (**few-shot classification**) to evaluate whether it can classify genres more effectively than traditional methods.

Discrepancies between predicted and actual genres are a central focus of the analysis, as they may reveal genre overlap, ambiguity, or evolving musical conventions. Even when predictions do not match the labeled genre, these differences provide meaningful insight into how genres are defined and interpreted.

An optional extension includes a time-series analysis to study how genre characteristics change over time.

What's in a name? That which we call a rose
By any other name would smell as sweet
- Romeo and Juliet, Shakespeare